

What's the energy impact of AI? I searched the real numbers , and created a video (with AI ;) to clarify the issue, the risks and the perspectives

Posted by u/Far-Key-8874 • 0 points • 11 comments

I believe new innovations will help reduce environmental impact. France is the most advanced in fusion energy, that will be one of the cleanest and more powerfull source of power

Comments

u/Murky-Opposite6464 • 2 points • 2025-10-01 02:02 UTC

What a coincidence! So did I!

The true effect of AI on the environment.

I've written about this before, but I felt like this topic deserves a deeper, more granular analysis of the environmental impact of AI. What I intend to do is show the current environmental impact of AI, compare that to the improvements AI has proven it is capable of, and come up with an estimate of how much damage/benefit is done by AI overall.

Since I am pro-AI myself, I'm going to endeavour to be fair by going by the highest figures I can find, even if those are estimates of future use.

First, the easy part, environmental damage.

"Data centers' electricity consumption in 2026 is projected to reach 1,000 terawatts (tWh), roughly Japan's total consumption".

That combined with the fact that AI is estimated to soon be using 50% of data centre power leaves is with 500 terawatts

"AI demand for water in 2027 would be 4.2-6.6 billion cubic meters, roughly equivalent to the consumption of 'half of the United Kingdom"

<https://begin.berkeley.edu/reducing-ais-climate-impact-everything-you-always-wanted-to-know-but-were-afraid-to-ask/>

I've excluded CO₂ emissions here because they are essentially a byproduct of energy use. Since the carbon intensity of electricity generation varies across regions, I chose to focus directly on the raw energy numbers, with the understanding that reductions in energy use will translate to a reduction in CO₂.

So, here are the scores.

Electricity use: 500 terawatt hours

Water use: 6.6 billion cubic meters

These are the numbers I will be comparing positive environmental impact with. And since I started writing this section first, I'm just as excited to see what the results will be as you are. :)

I'm going to be going through this field by field, sector by sector, and totalling up the results I find. To ensure total fairness, any range I am given, I will be using the lowest estimate.

Let's start with agriculture.

We are currently using 4.3 trillion cubic meters of fresh water per year, with 70% of that going to agriculture.[1] That means we

are using 3 trillion cubic meters just for growing food.

I found many reports on AI controlled irrigation techniques reducing water usage by up to 30%, increasing crop yields by 20% at the same time. However, these were in third world countries that likely aren't using the most efficient non-AI methods to begin with. For a fairer comparison, I looked to a test performed in Canada that resulted in between 6.4% and 22.8% in water reduction, and a 2.3% to 4.3% increase in yield. So, going with my lowest estimate policy, that's a reduction of 6.4%, plus a negligible amount for increased crop yields[2].

If we were to apply this system globally, that's a 192 billion cubic meter reduction in water use. It can reduce herbicide use too by using AI targeting, but we won't factor that in.

Well... we sure past that 6.6 billion cubic meters of water lost metric quickly.

Let's move on to the next sector. Data centres!

The data centres that power AI, and the site you are reading this on, have been a big topic as far as increases in electrical use. However, little is said about how AI can make those data centres more efficient.

Google claimed that by using their Deepmind AI, they cut the cost of cooling their data centres by "up to" 40%, 15% of the data centres total power use.

I saw many claims of similar numbers, usually around 30%. The lowest I could find is a reduction in cooling costs from between 14% and 21%[3]. So, let's go with 14%.

If Google's 40% claim lead to a 15% reduction overall (which seems to match other sources), a 10% reduction on cooling should be a 5.25% total reduction. So 52.5 terawatt hours.

Air conditioning!

When it comes to temperature control in general for places like offices and department stores, I found cuts of up to 67% in various studies. The lowest I could find was 24.52%[4]. Since it had a variance of 10%, we'll go with 14%.

To get the air conditioning power use for only large buildings, I'm simply going to subtract residential air conditioning energy use from total air conditioning use.

We get 2,100 terawatt hours for total air conditioning[5], and residential air conditioning units are 1,200 terawatt hours[6], so 900 terawatt hours for more industrial air conditioning. That's a reduction of 126 terawatt hours.

Solar panel and wind farm optimization!

This one was fairly straightforward. AI can help position solar panels to increase energy production by 20% (I saw claims as high as 30%), and wind farm power output by 12%[7].

Solar produces 1,600 terawatt hours per year[8], leading to an increase of 320 terawatt hours. Wind generates 2,330 terawatt hours[9], so an increase of 279 terawatt hours.

So, even by the lowest metrics I could find, the increased efficiency for wind and solar farms on its own would basically offset AI's energy consumption entirely, making everything else I've cited an actual reduction in global energy use. All thanks to AI.

I hope this will counter some of the misinformation I've been seeing out there, and put some concerns to rest. There will be challenges that come with the growth of AI, a lot will change. However, as far as the environment is concerned, these changes will be for the better.

1: https://reports.weforum.org/docs/WEF_Water_Futures_Mobilizing_Multi_Stakeholder_Action_for_Resilience_2025.pdf

2: <https://arxiv.org/abs/2306.08715>

3: <https://arxiv.org/abs/2501.15085>

4: https://www.researchgate.net/figure/The-average-energy-savings-of-the-24-cases-and-the-maximum-energy-savings-achieved-by_fig6_331608687

5: <https://ourworldindata.org/air-conditioning-causes-around-greenhouse-gas-emissions-will-change-future>

6: <https://www.nature.com/articles/s41467-024-52028-8>

7: https://ojs.stanford.edu/ojs/index.php/intersect/article/view/3541?utm_source=chatgpt.com

8: <https://www.iea.org/energy-system/renewables/solar-pv>

9: <https://www.iea.org/energy-system/renewables/wind>

Fun little bonus facts. The positive effects of AI on carbon emissions in just 3 industries (meat and dairy, power, and light road vehicles) would completely negate all damage done by increased power consumption from expanding AI use.

<https://www.nature.com/articles/s44168-025-00252-3>

UPS optimized their deliveries with AI, saving 10 million gallons of fuel.

https://www.bestpractice.ai/ai-case-study-best-practice/ups_saves_over_10_million_gallons_of_fuel_and_up_to_%24400m_in_costs_annually_with_advanced_telematics_and_analysis

Pittsburg used AI controlled stoplights to cut emissions from cars by 21%, and reduced total driving time by 26%.

<https://govlaunch.com/stories/pittsburgh-pa-reduces-traffic-congestion-with-ai>

u/Ok_Butterfly1799 • 1 points • 2025-09-30 21:51 UTC

ai data centers are using clean,drinking water from towns and polluting it

u/ZorotheGallade • 0 points • 2025-09-30 18:34 UTC

Frankly? Waiting for "new technologies" to bail us out on the problems we create with the current ones is a very dangerous line of thought.

u/TheCyanHoodie • 2 points • 2025-09-30 21:34 UTC

No, no you don't get it, well just keep going harder and harder inventing newer and better things without actually making sure our power sources aren't irreversibly harming our planet and destroying the lives of people by up running their electricity bill! I'm SURE someone is this close 🍌 to inventing a nuclear-fusion-hydrogen-clean-buzzword-hydro energy source that will solve everything!!!

u/Dmayak • 2 points • 2025-09-30 17:27 UTC

Why do a lot of people just assume that any kind of development trend will just continue infinitely? Any "hot new thing" in demand has a phase of booming supply growth until it goes above demand and then it goes down and reaches some sort of equilibrium. Electricity, maintenance, cooling and all that crap isn't free, and even large companies can't just spend on it endlessly. Especially if it will become more in demand and prices on it will grow.

u/Flashy_Cranberry_161 • 0 points • 2025-09-30 17:12 UTC

"Ermmm it's actually not a problem because other things consumes more energy than AI huurrrr durrrr"

u/Situati0nist • 8 points • 2025-09-30 17:36 UTC

It is a problem but so are a lot of other things. Hyperfixating on AI being *the* problem for the environment is shortsighted no matter how you slice it.

u/Flashy_Cranberry_161 • 1 points • 2025-09-30 17:54 UTC

This is what they said when it was revealed crypto mining was a big energy suck to. And I say the same thing now as I did back then.

"Stupid waste of power in a time when we need to think about energy efficiency or downsizing"

u/Situati0nist • 2 points • 2025-09-30 19:14 UTC

Right, but only cutting everything you'd deem unnecessary is not a viable solution. Not only is AI not going to go away, the supposed idea is not enough. We need a much more effective and sustainable way to power everything without too many emissions, zero if possible.

I wish society would focus more on nuclear energy. I don't know what is happening but they're

shutting down these power plants rather than building more, which is a shame seeing how it has a few downsides and mostly tremendous upsides.

u/CalicoCatio • 1 points • 2025-09-30 20:07 UTC

[They are](<https://www.npr.org/2024/09/20/nx-s1-5120581/three-mile-island-nuclear-power-plant-microsoft-ai>)

u/Flashy_Cranberry_161 • 1 points • 2025-09-30 21:53 UTC

Jesus christ. All of this for AI contractors and not the American people. Really makes you wonder about this countries priorities